

Quiz-06-AMBIG-PREC

Quiz Q1:

A production rule consists of:

- ☐ Terminals
- ☐ Regular Expressions
- ☐ Non-terminals
- ☐ Function calls

Quiz Q2:

a left derivation will always produce the same parse tree as a right derivation

☐ True

☐ False

Quiz Q3:

What is an example of input recognized by the following grammar?

$a \rightarrow aX$

$a \rightarrow Y$

☐ XXXXXXXXXY

☐ XYYYYYYYYY

☐ YXXXXXXXXX

☐ YYYYYYYYX

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

XXXXXXXXXY

RULE	Sentential Form
start	a

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

XXXXXXXXY

RULE	Sentential Form
start	a

*Applying either rule
gives us a sentential
form that won't create
the string*

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

XYYYY

RULE	Sentential Form
start	A

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

XYYY

RULE	Sentential Form
start	A

Similar reason:
strings that are
longer than 1 character
cannot end in y

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

YXXXXXX

RULE	Sentential Form
start	A

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

YXXXXXX

RULE	Sentential Form
start	A
1	aX
1	aXX
...	
2	YXXXXXX

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

YYYYYYX

RULE	Sentential Form
start	A

Quiz

What is an example of input recognized by the following grammar?

- 1 $a \rightarrow aX$
- 2 $a \rightarrow Y$

How about this one?

YYYYYYX

We can only produce
1 y, so we cannot
derive this string

RULE	Sentential Form
start	A

Quiz Q4:

Which grammar is ambiguous?

(a)

$e \rightarrow e \text{ PLUS } e$

$e \rightarrow \text{ID}$

(b)

$e \rightarrow e \text{ PLUS ID}$

$e \rightarrow \text{ID}$

(c)

$e \rightarrow \text{ID PLUS } e$

$e \rightarrow \text{ID}$

(d)

$e \rightarrow \text{ID PLUS ID}$

$e \rightarrow \text{ID}$

Quiz

Which grammar is ambiguous?

(a)

$e \rightarrow e \text{ PLUS } e$

$e \rightarrow \text{ID}$

(b)

$e \rightarrow e \text{ PLUS ID}$

$e \rightarrow \text{ID}$

(c)

$e \rightarrow \text{ID PLUS } e$

$e \rightarrow \text{ID}$

(d)

$e \rightarrow \text{ID PLUS ID}$

$e \rightarrow \text{ID}$

Let's look at some examples.

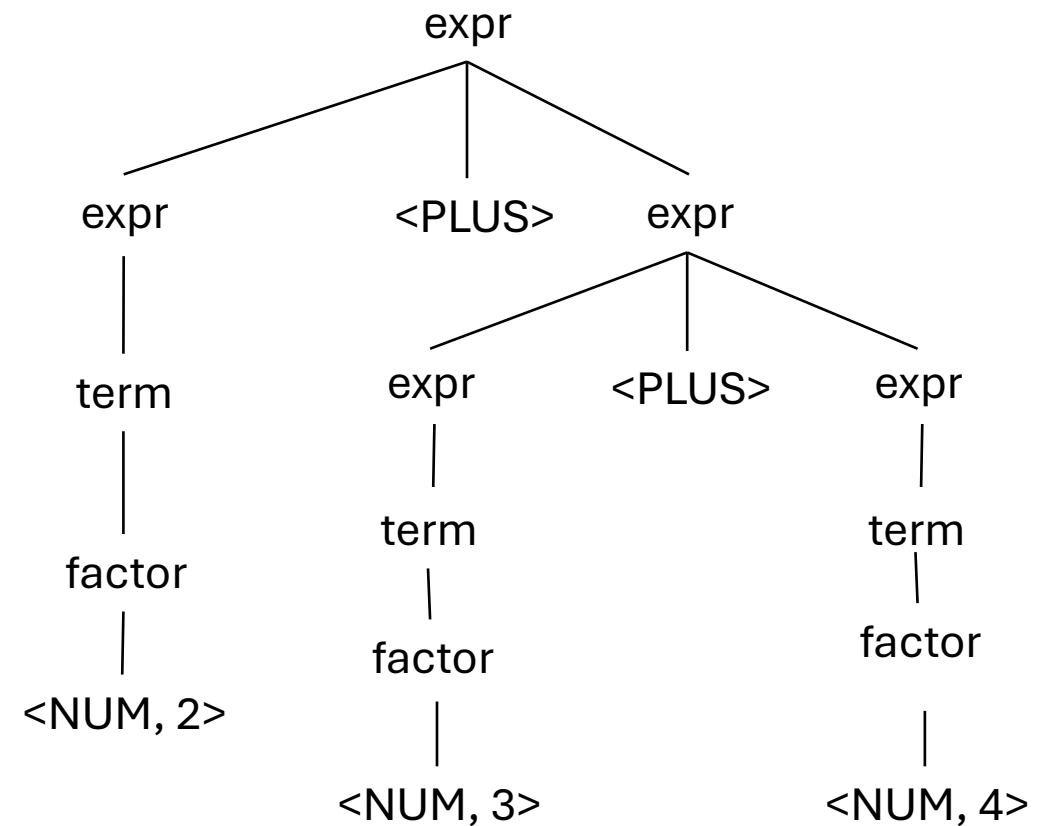
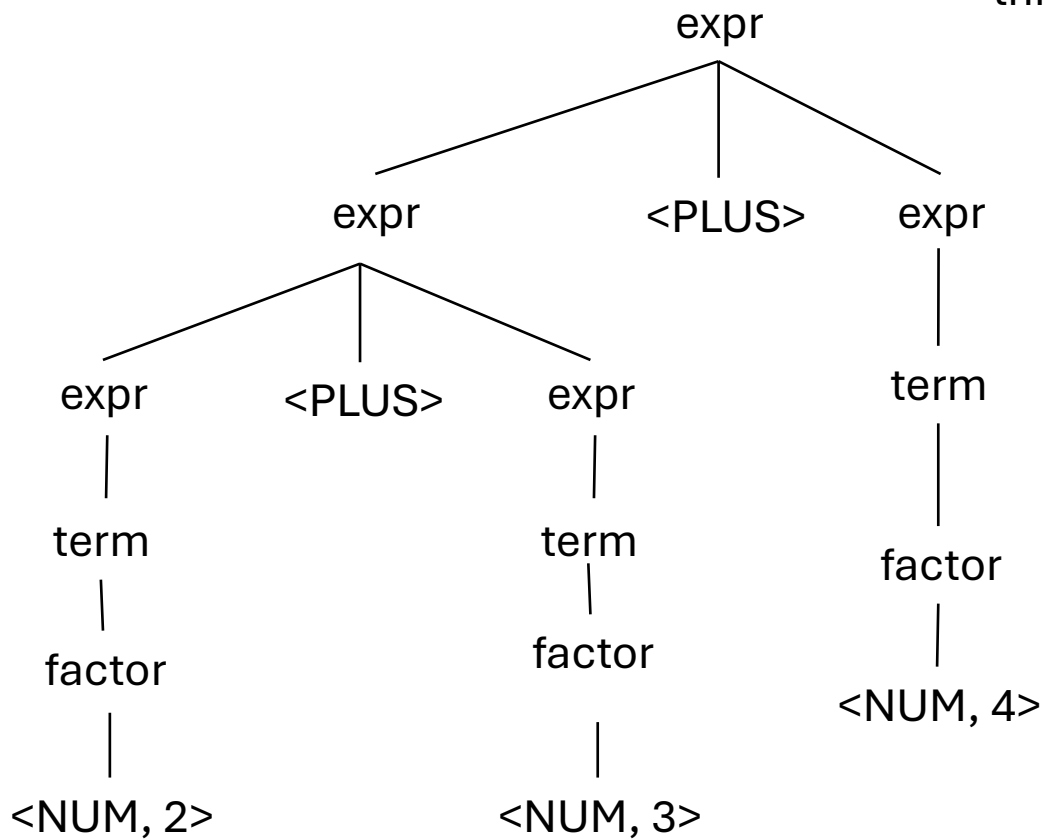
Let's assume that E is an "expr"
and x is a number

$e \rightarrow e \text{ PLUS } e$

$e \rightarrow \text{ID}$

input: 2+3+4

Both parse trees are valid,
this grammar is ambiguous



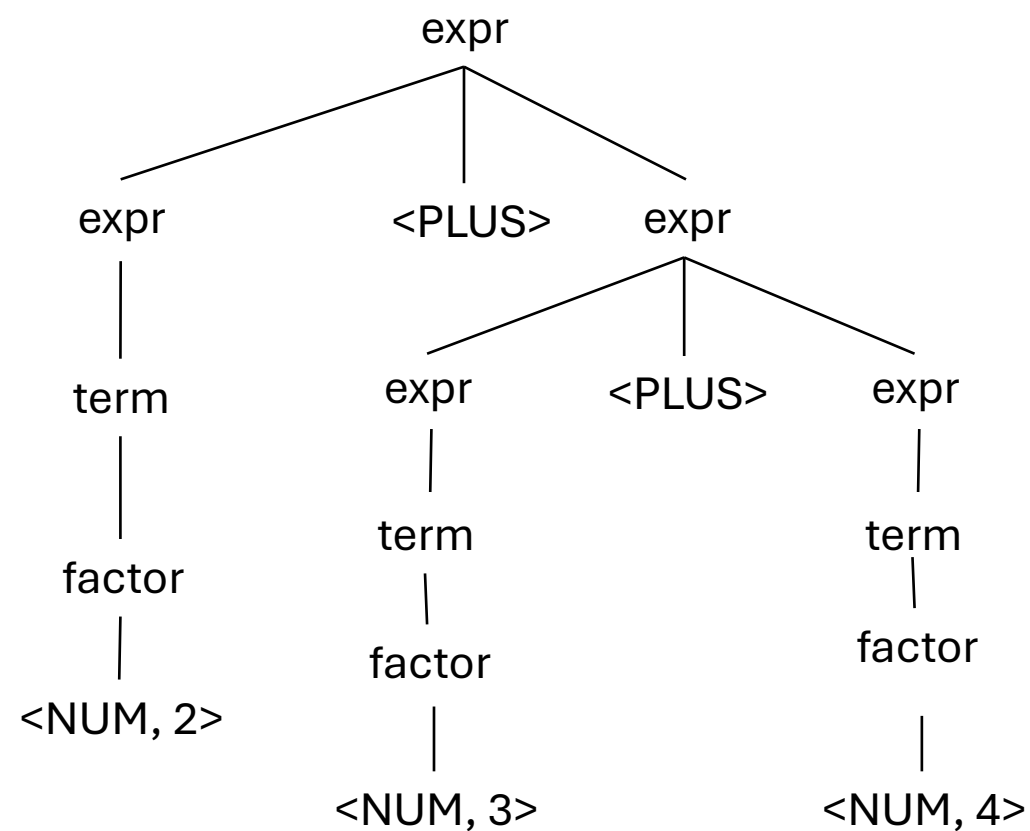
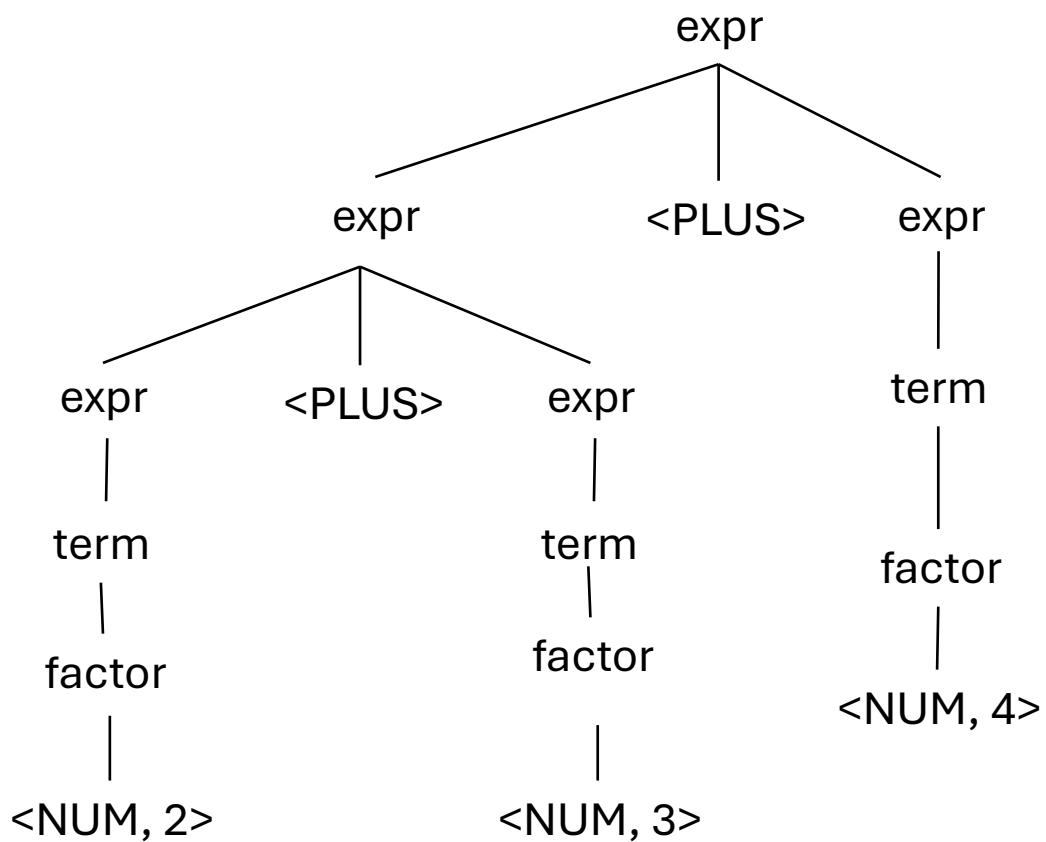
$e \rightarrow e \text{ PLUS ID}$

$e \rightarrow \text{ID}$

input: 2+3+4

See next slide

What about this one?

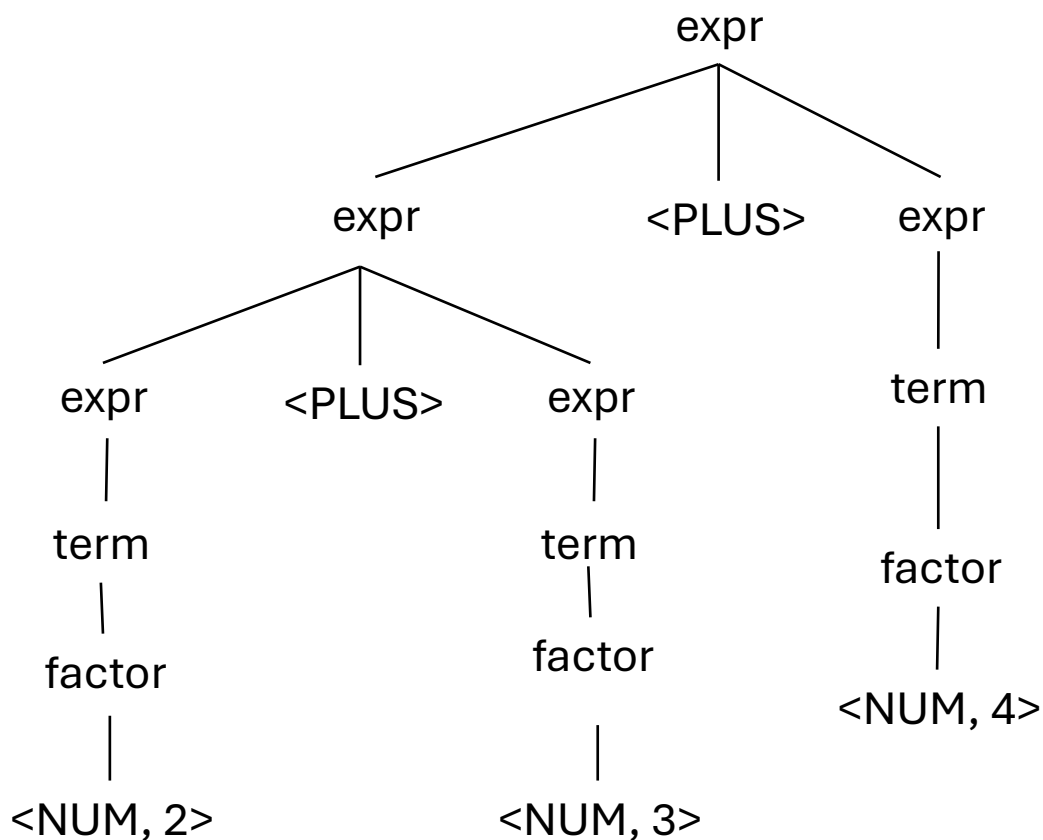


$e \rightarrow e \text{ PLUS ID}$
 $e \rightarrow \text{ID}$

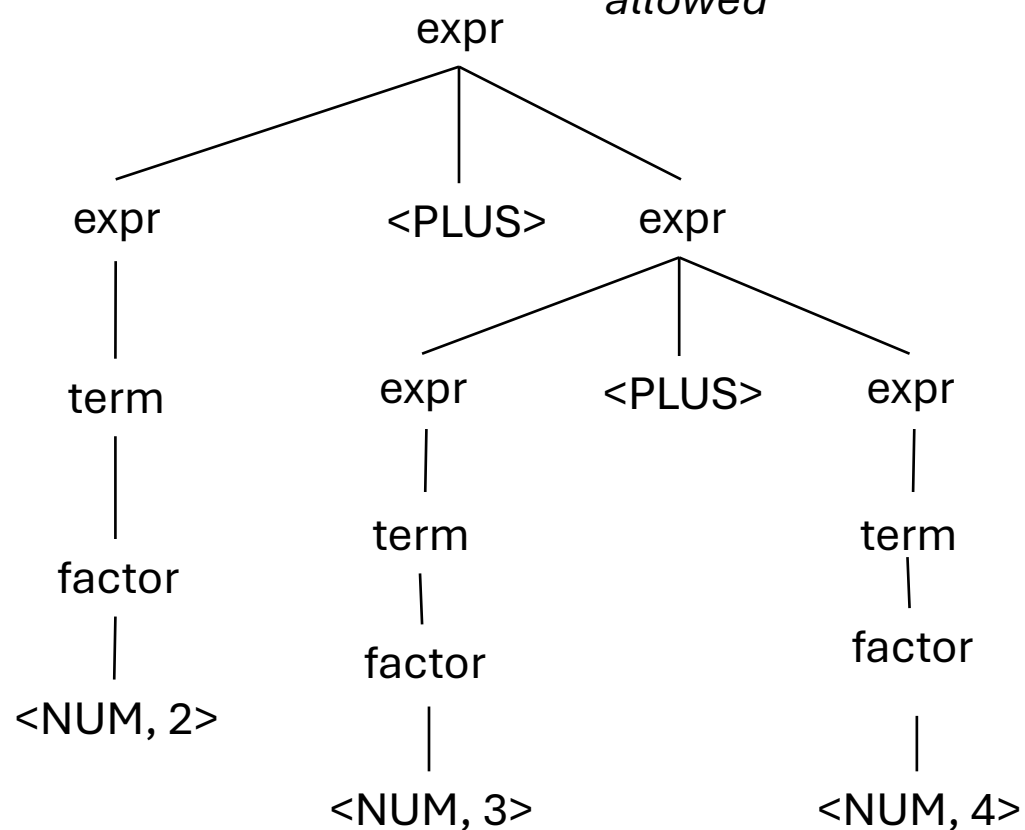
input: 2+3+4

UNAMBIGUOUS

What about this one?



Doesn't allow an expression on the RHS. This parse tree is not allowed



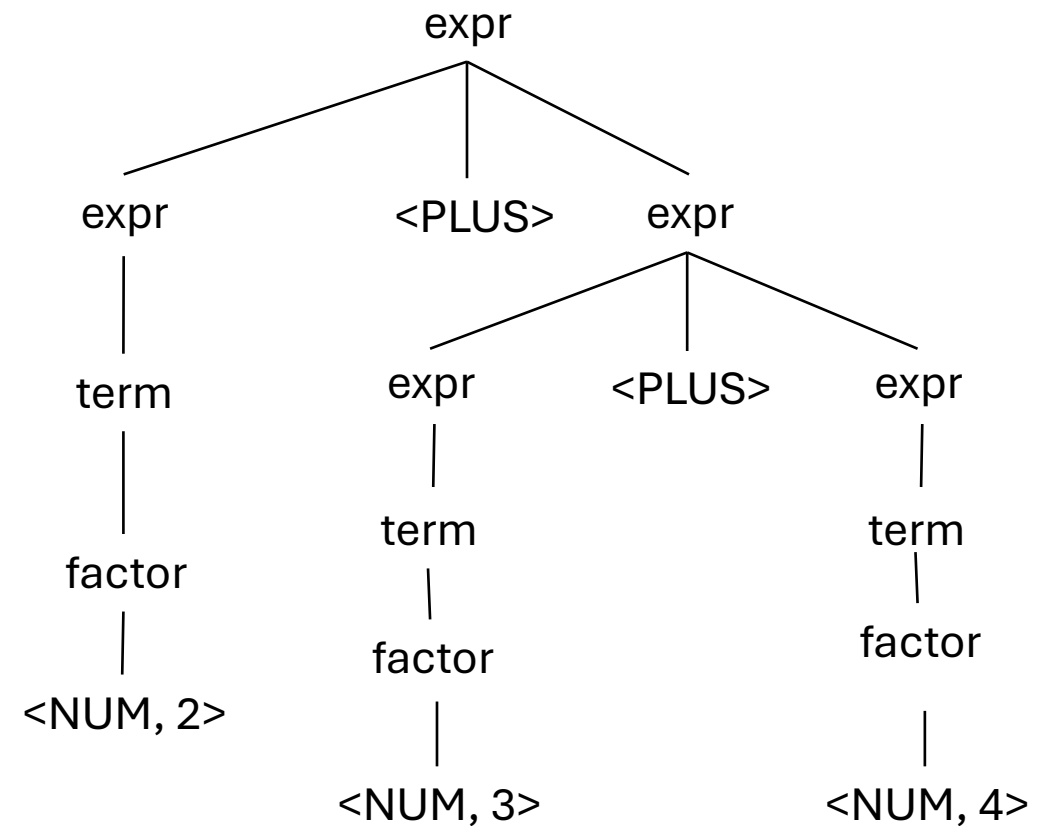
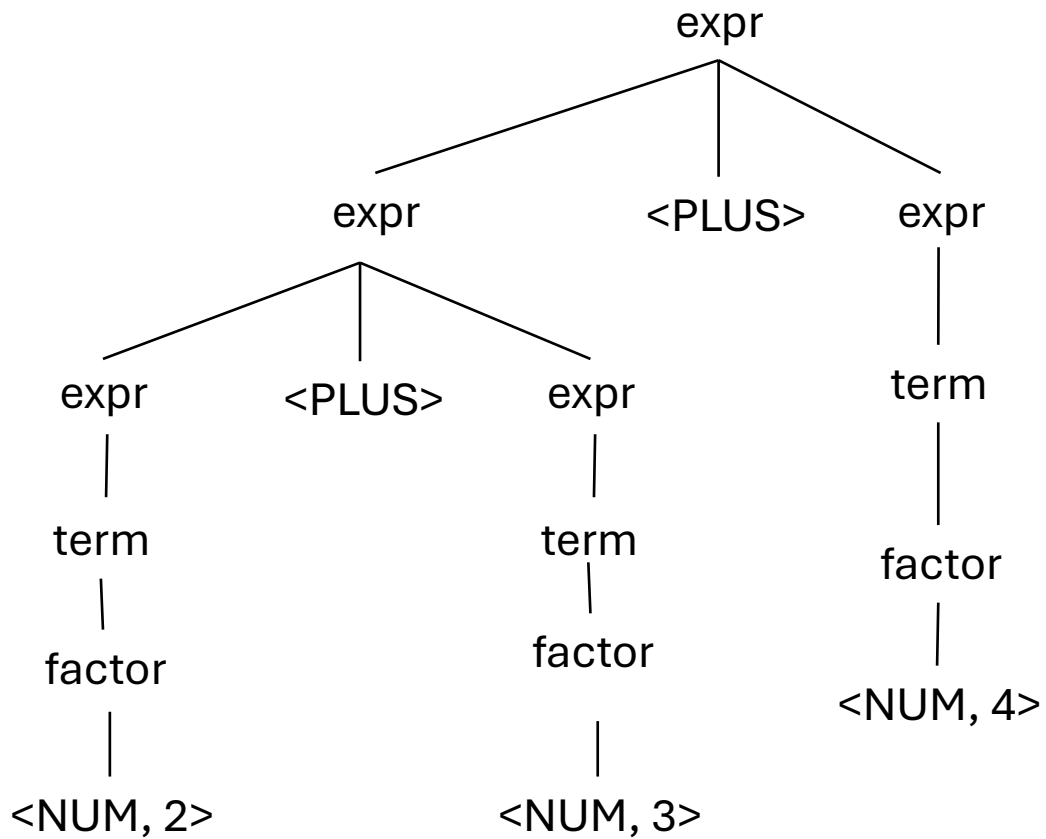
$e \rightarrow \text{ID PLUS } e$

$e \rightarrow \text{ID}$

input: 2+3+4

See next slide

What about this one?



$e \rightarrow \text{ID PLUS } e$

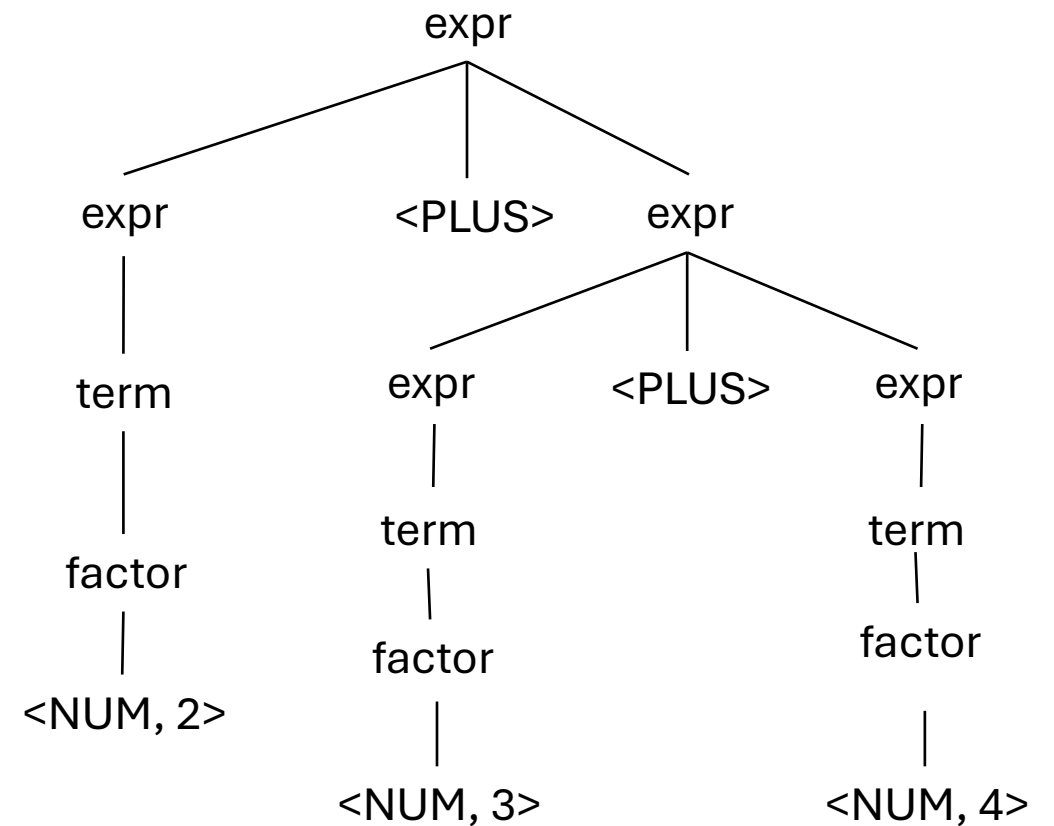
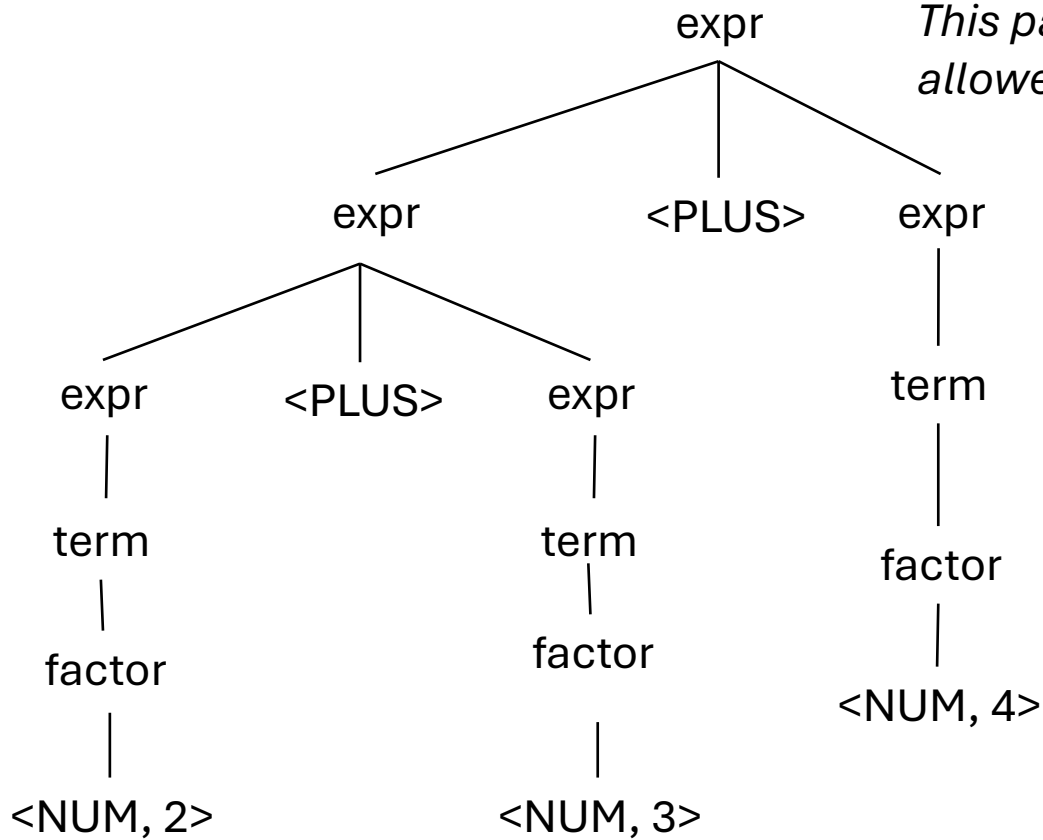
$e \rightarrow \text{ID}$

input: 2+3+4

UNAMBIGUOUS

What about this one?

*Doesn't allow an
expression on the LHS.
This parse tree is not
allowed*



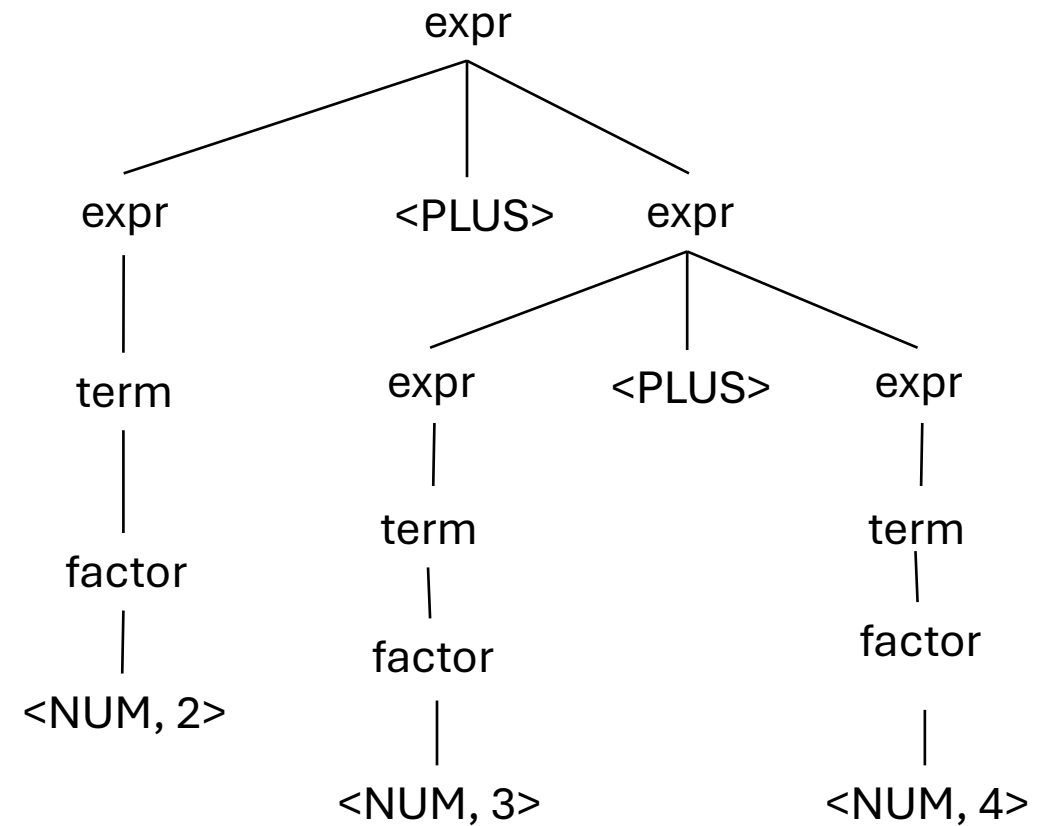
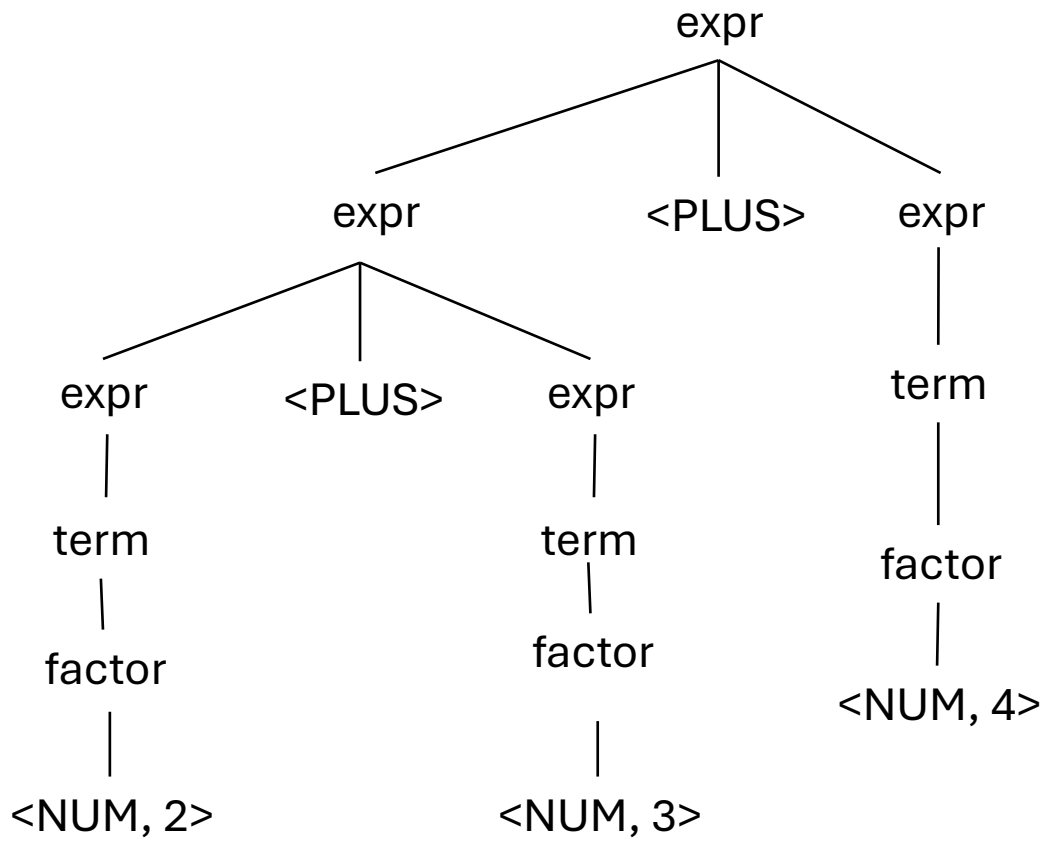
$e \rightarrow \text{ID PLUS ID}$

$e \rightarrow \text{ID}$

input: 2+3+4

See next slide

What about this one?



$e \rightarrow \text{ID PLUS ID}$

$e \rightarrow \text{ID}$

input: 2+3+4

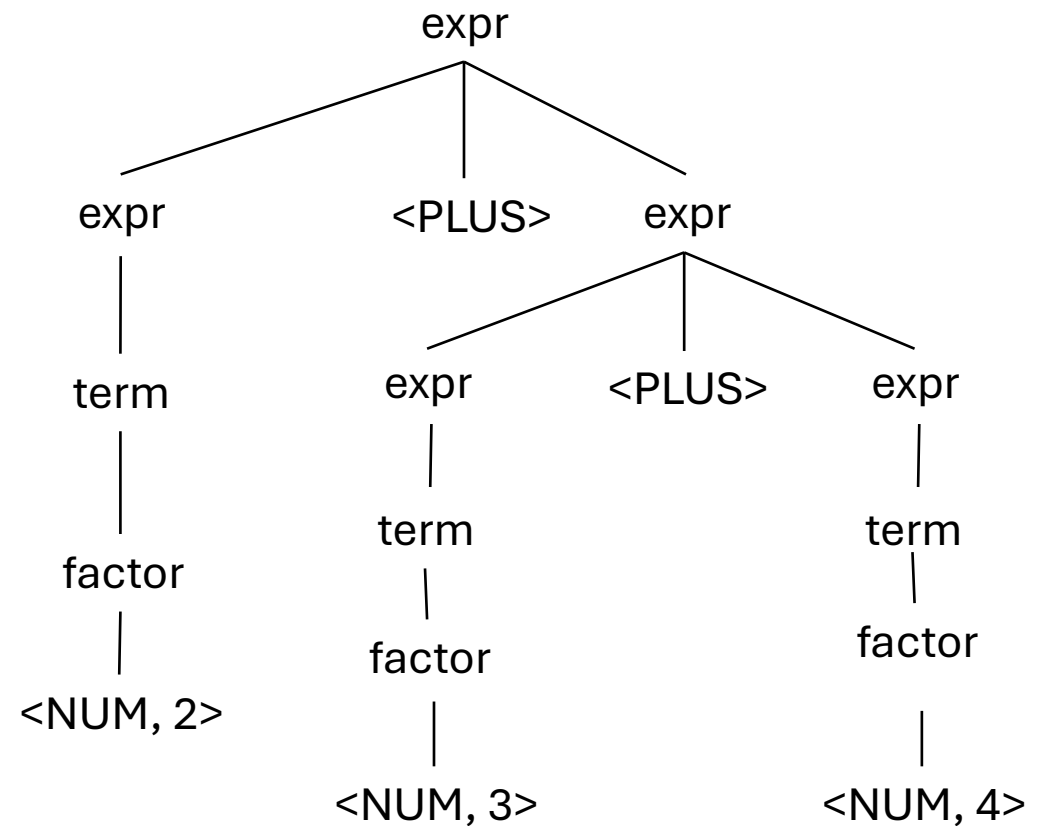
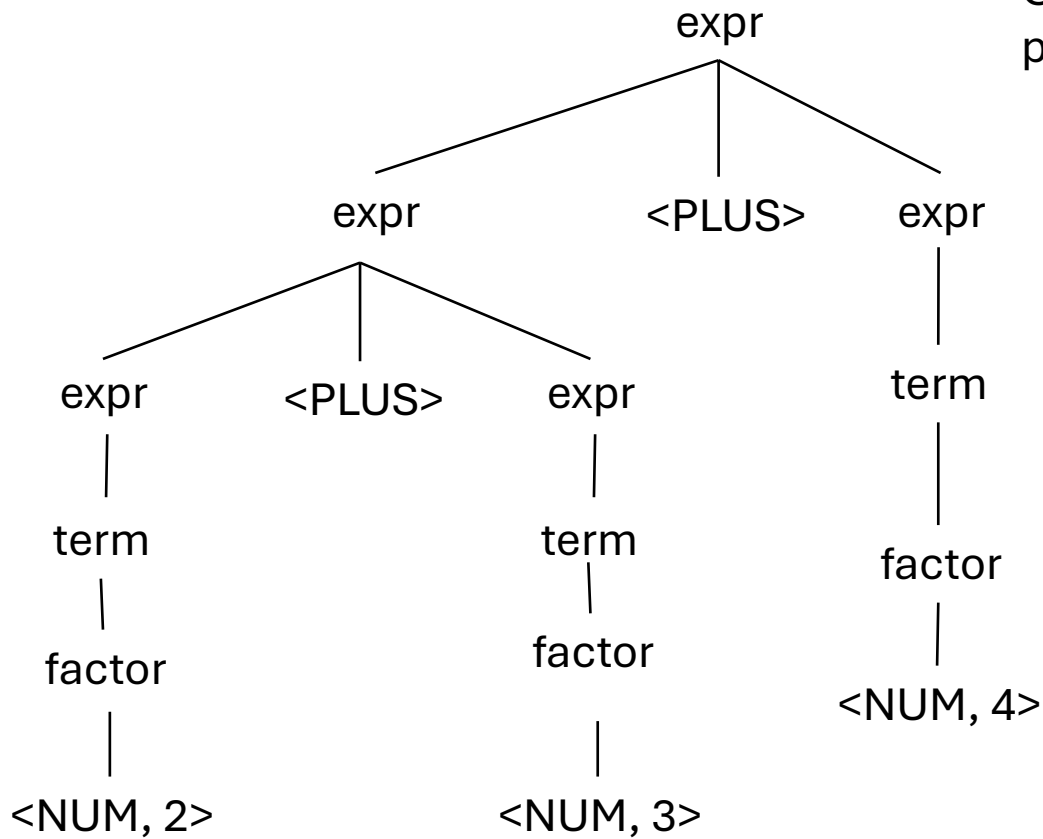
UNAMBIGUOUS

But not expressive

No recursion

What about this one?

Cannot produce either
parse tree!



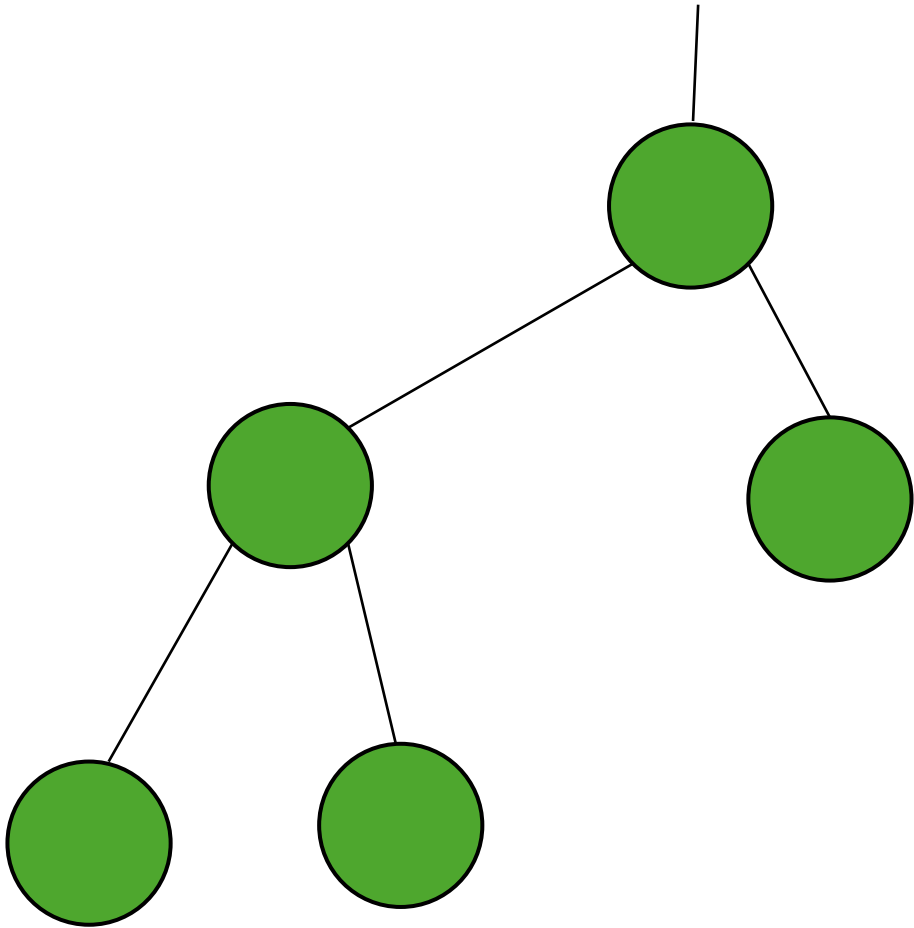
Quiz Q5:

operators with higher precedence should appear in production rules that appear higher in the parse tree

☐ True

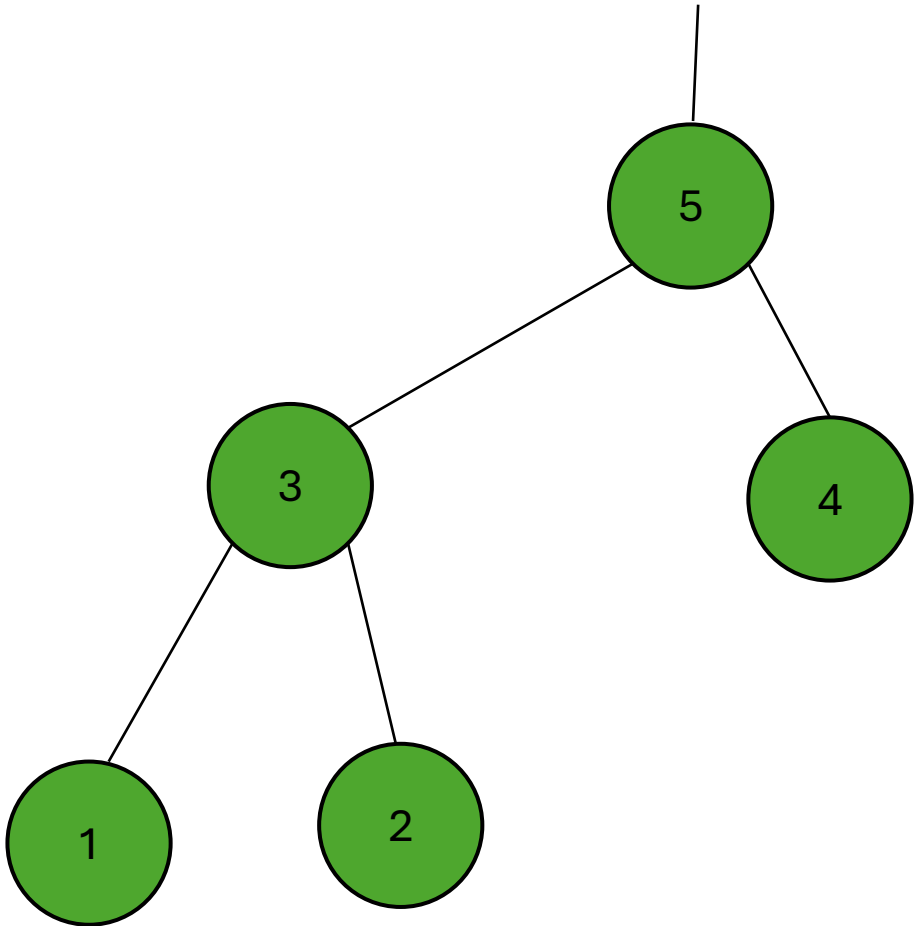
☐ False

Post order traversal



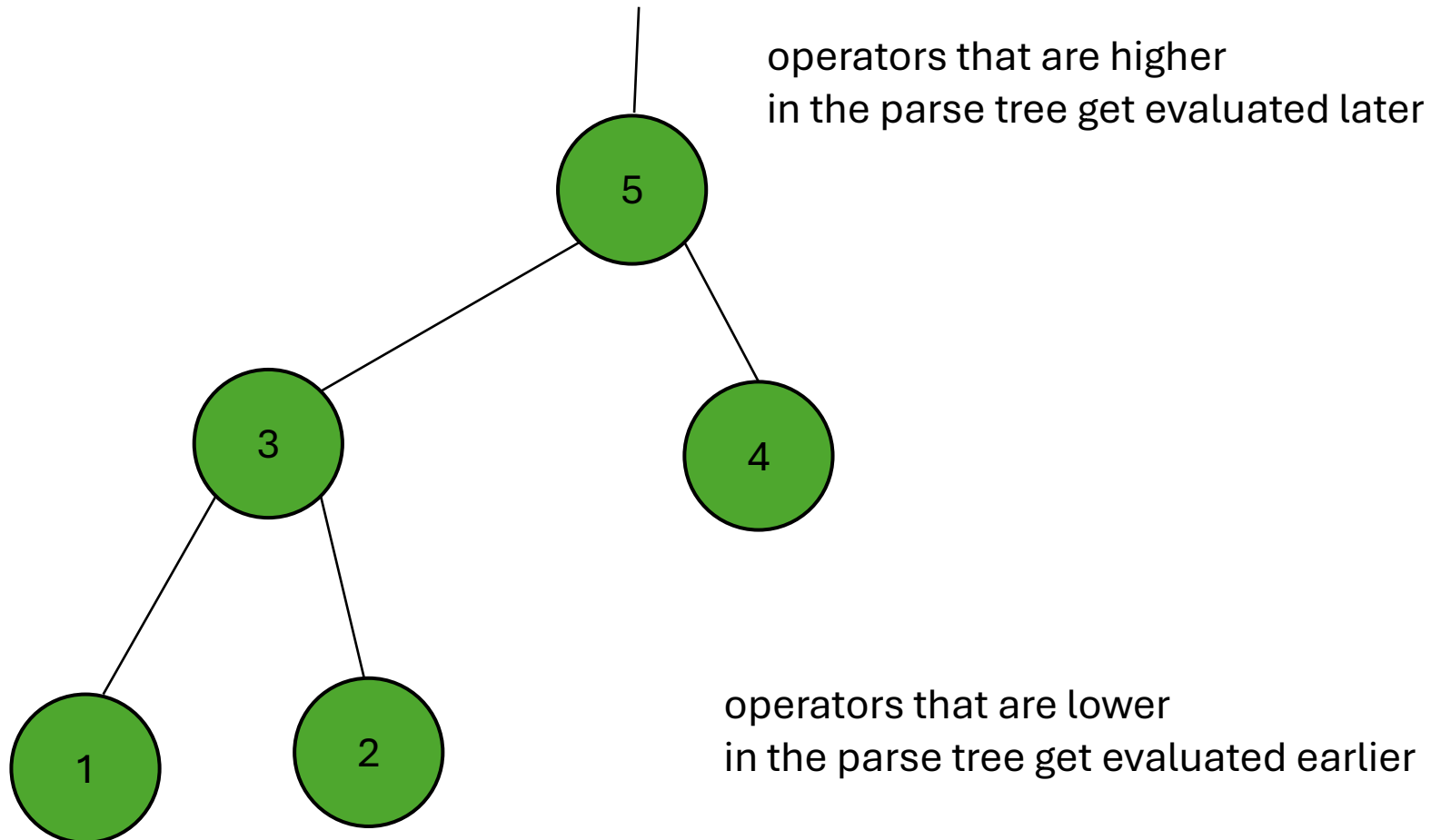
What is the post order traversal of this tree?

Post order traversal



What is the post order traversal of this tree?

Post order traversal

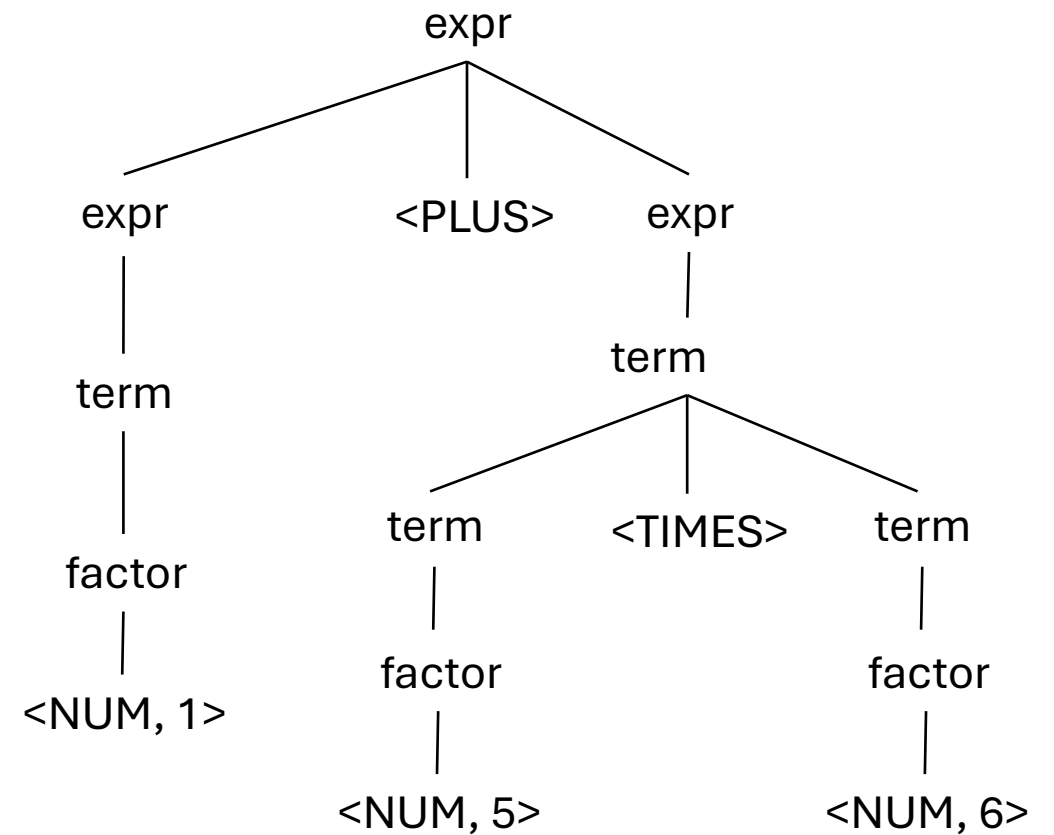


What is the post order traversal
of this tree?

Evaluating a parse tree

input: 1+5*6

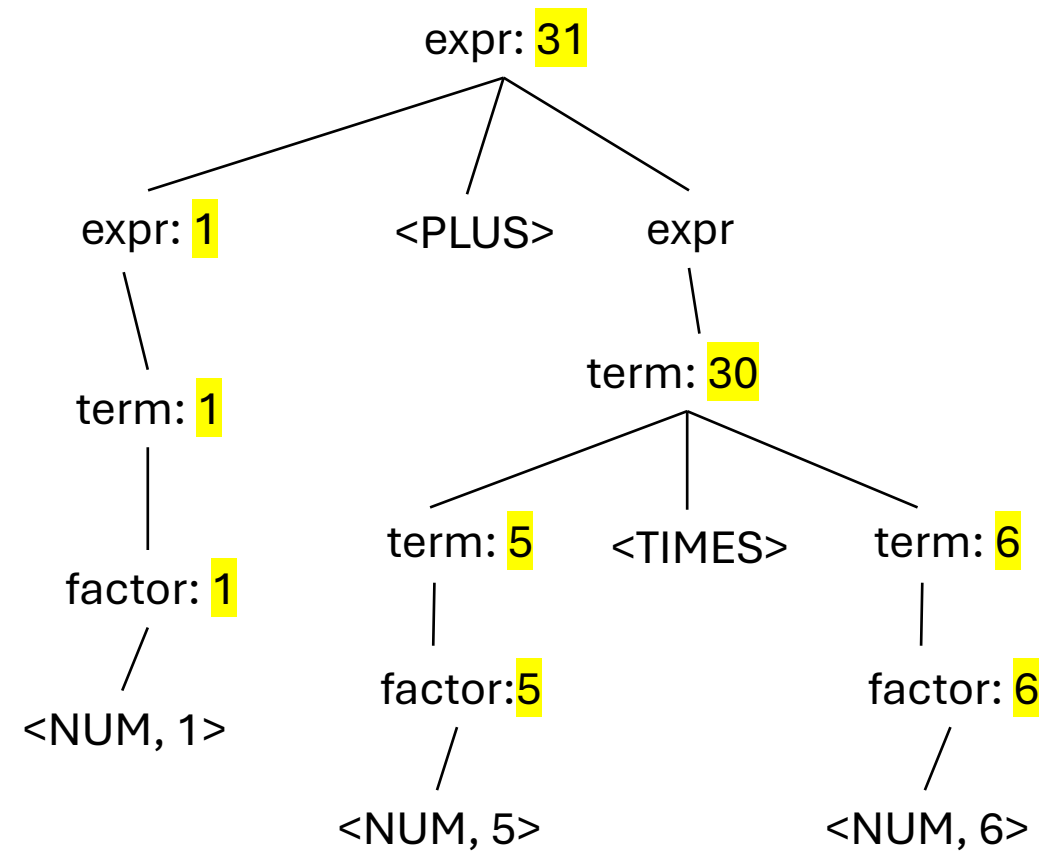
Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Evaluating a parse tree

input: 1+5*6

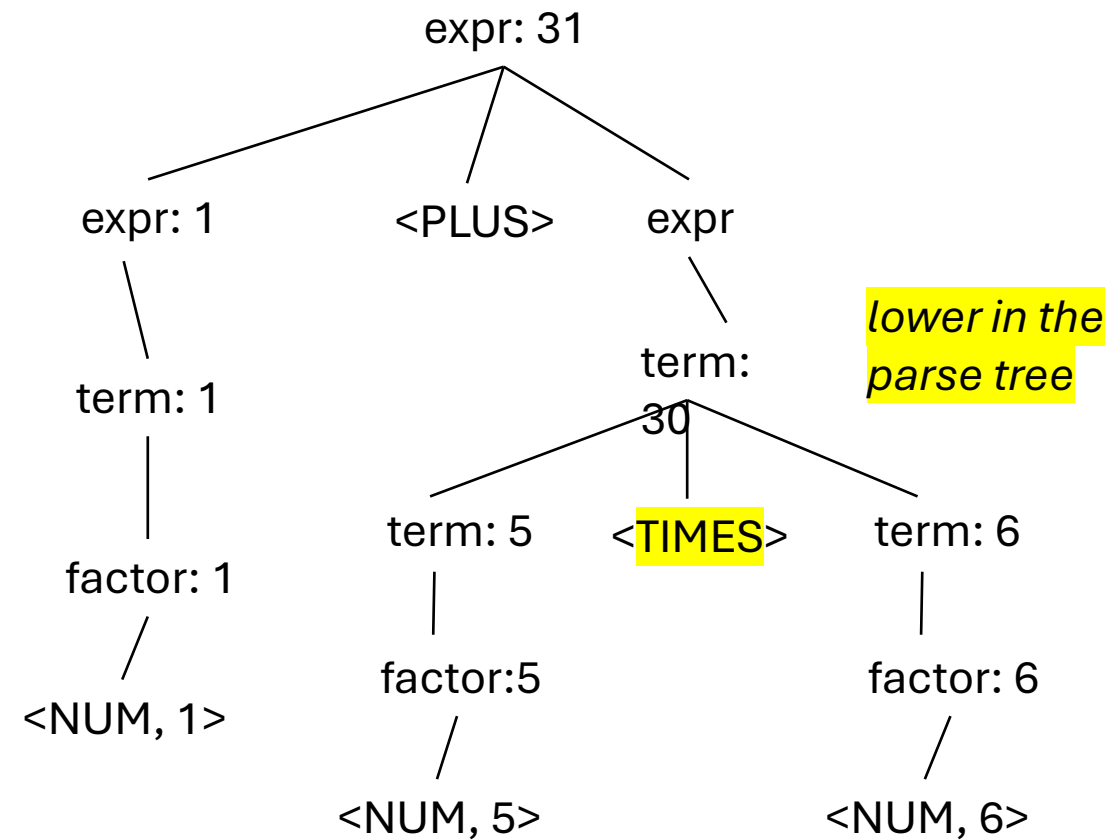
Operator	Name	Productions
+	expr	: expr PLUS expr term
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Evaluating a parse tree

input: 1+5*6

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Avoiding Ambiguity

- new production rules
 - One non-terminal for each level of precedence
 - lowest precedence at the top
 - highest precedence at the bottom
- How would we add power? ^

See this: [See this](#)

Precedence
increases going down

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



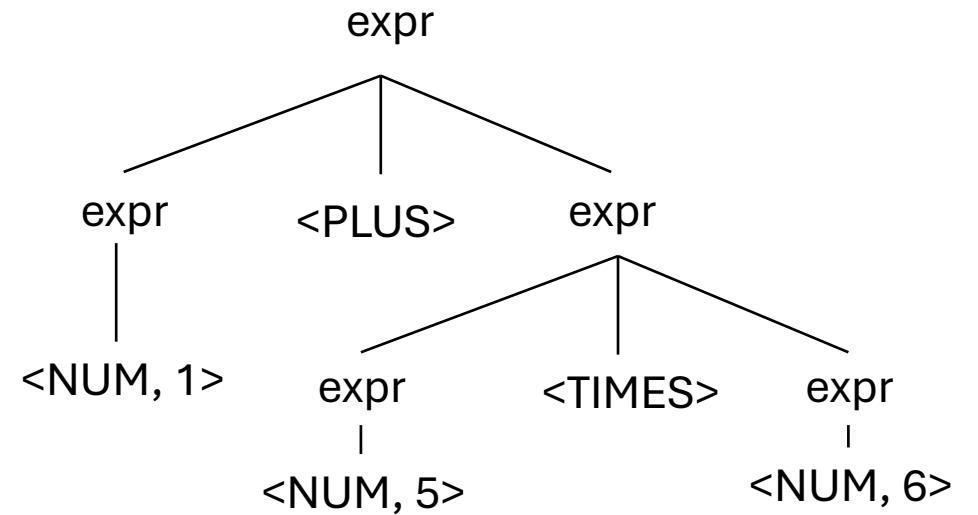
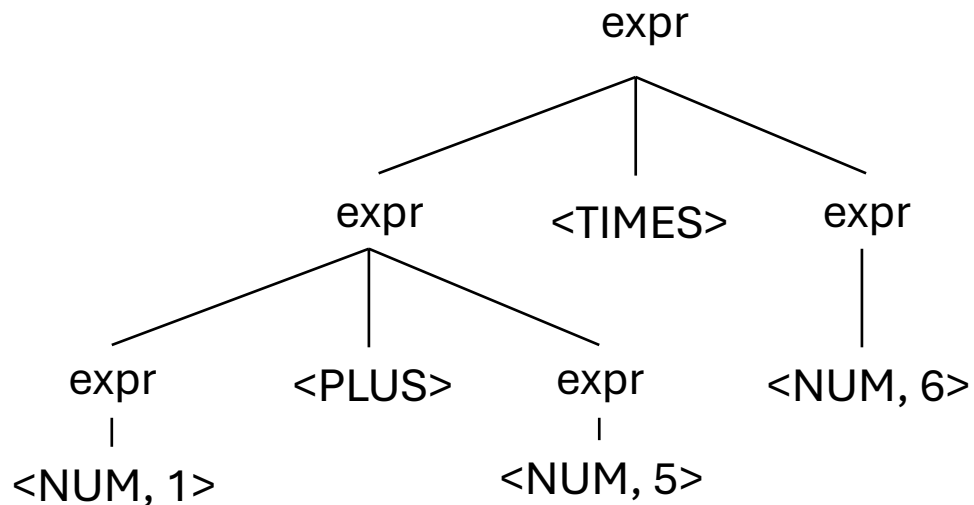
Quiz Q6:

Write a few sentences about why it might be bad to have an ambiguous grammars

Ambiguous grammars

- input: 1 + 5 * 6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Ambiguous grammars

- input: 1 + 5 * 6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

Evaluations are different!

